

To find following the ML regression method using in r2 value

1. Multiple Linear Regression:

R2 value = 0.9358

2. Support Vector Machine:

S.NO	Hyper Tuning Parameter	Linear (r value)	RBF (Non Linear) (r value)	Poly (r value)	Sigmoid (r value)
1	C=10	-0.13	-380533.40	-495.811	-35675016.60
2	C=100	-1.29	-4265.22	-2.66	-357142.02
3	C=500	-1.41	-212.78	-0.10	-14355.37
4	C=1000	Consume more time to run	-46.22	-0.11	-3610.64
5	C=2000	Consume more time to run	-10.98	-0.04	-913.57
6	C=3000	Consume more time to run	-4.63	0.005	-410.89

The SVM Regression use **R2 Value** (poly and hyper parameter (C=3000)) = **0.005**

3. Decision Tree:

S.NO	Criterion	Max Features	Splitter	R Value
1	squared_error	None	best	0.8917
2	squared_error	None	random	0.8903
3	squared_error	sqrt	best	0.5467
4	squared_error	sqrt	random	0.5819
5	squared_error	log2	best	0.4717
6	squared_error	log2	random	0.2025
7	absolute_error	None	best	0.9308
8	absolute_error	None	random	0.8953
9	absolute_error	sqrt	best	0.7957
10	absolute_error	sqrt	random	0.9131
11	absolute_error	log2	best	0.8742
12	absolute_error	log2	random	0.7173
13	poisson	None	best	0.9324
14	poisson	None	random	0.9497
15	poisson	sqrt	best	-0.9597
16	poisson	sqrt	random	0.3318
17	poisson	log2	best	-0.3873
18	poisson	log2	random	0.7543

The Decision Tree Regression use **R² Value**
(poisson, None, random) = **0.9497**